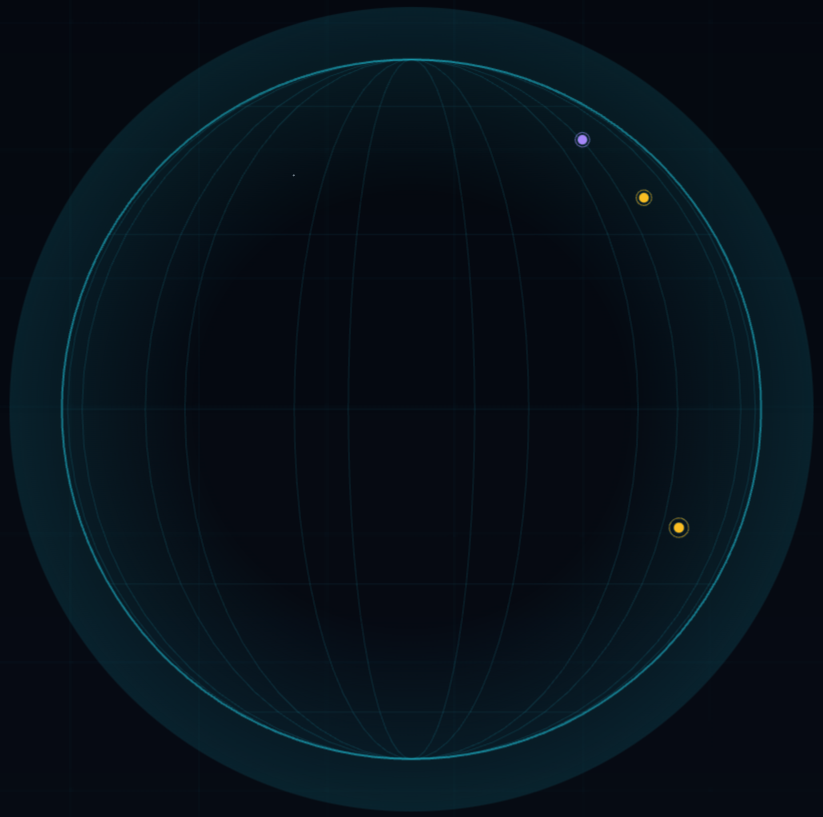
DECISION SIMULATION FOR CRISIS LEADERS

DASHBOARDS SHOW THE CRISIS. WE SIMULATE YOUR DECISION.

A holographic AI war room that ingests live disaster data, forecasts escalation on the GPU, and lays three grounded response options on a 3D Earth, with honest ranges instead of false precision.

AMD GPU | ROCM · FIREWORKS AI | PYTORCH MONTE CARLO



THE GAP

Everyone shows the crisis. Nobody simulates the response.

TODAY: SITUATIONAL AWARENESS

WHAT IS HAPPENING

Crisis24 and Dataminr, public dashboards like GDACS and ReliefWeb all detect and display events in real time. The feed tells you a flood is rising.

THE MISSING LAYER

WHAT HAPPENS NEXT, UNDER EACH CHOICE

What is exposed in 6 / 24 / 72 hours if nothing is done? Evacuate now, pre-position, or monitor: at what cost, averting whom? **No open-source tool answers this.**

The decision layer is the hard part. It is the part that needs simulation, grounding, and a GPU. That is exactly what CrisisCommand builds.

THE DEMO — 60 SECONDS

One globe. Three modes. A decision.

01 **Live Earth.** GDACS + USGS events pulse on a holographic globe, severity-colored, ingested minutes ago.

02 **Click a crisis.** The camera flies in; an AI situation brief streams in word by word.

03 **Run simulation.** 10,000 Monte Carlo runs on the AMD GPU; the readout names the card and spikes; the escalation curve draws itself.

04 **Three options.** Evacuate / pre-position / monitor. Each carries an exposed-population range, cost, response time, and honest trade-offs.

05 **Choose.** Selecting an option renders its evacuation zones and supply arcs on the Earth in 3D.

ESCALATION FORECAST · JAKARTA FLOOD · 24H

52K–335K exposed · p10–p90

Every figure is a band. The tensor engine quantifies; the model never emits a single hard number.

SIMULATION · DECISION SUPPORT ONLY

—— WHY IT NEEDS THE GPU — ENGINEERED IN, NOT BOLTED ON

Three compute paths on one AMD GPU.

01 · SELF-HOSTED LLM

SCENARIO AGENT

The three policy branches are one batched LLM request. This demo runs them on **Fireworks** (AMD-powered); the same OpenAI-compatible client points at vLLM on the GPU with one env var: `SIM_BACKEND=vllm`.

02 · MONTE CARLO ENGINE

PYTORCH / ROCM

10,000 stochastic hazard-exposure runs as **one tensor batch**, no Python loops. Device-agnostic: ROCm presents as cuda, so the same code runs on any AMD GPU and on a laptop.

03 · EMBEDDING DEDUP

ON-GPU CLUSTERING

Incoming reports are embedded and cosine-matched on the GPU to merge duplicate events across feeds before they hit the globe.

MONTE CARLO · CPU BASELINE
519 ms · 100k runs

GPU SPEEDUP
61×
100k runs · measured on AMD gfx1100

VRAM ON THE CARD WE RAN
48 GB · gfx1100

GPU READOUT
live
visible during sim

Visible AMD usage is judged AMD usage. The utilization bar sits in the interface and spikes on every run. The speedup is measured on the AMD GPU (61× at 100k runs, 30.4M runs/sec), never estimated.

—— THE ETHICAL CORE — THIS IS A DECISION-SUPPORT TOOL

Honest by design.
Fake precision is disqualifying.

- **Every figure is a range.** p10–p90 bands, never a point estimate that pretends to certainty.
- **The LLM never invents numbers.** The tensor engine quantifies; the model only narrates and applies vetted mitigation factors to the ranges it is handed.
- **Everything is labeled.** All simulated data carries “SIMULATION · DECISION SUPPORT ONLY.”
- **A human decides.** Options with explicit trade-offs, never “the AI decided.” Scope is disaster response, never offensive action.

RESILIENCE: REHEARSED, NOT THEORETICAL

- Dead conference Wi-Fi → full offline demo on curated historical events
- LLM endpoint down → auto-fallback, honest banner in the HUD
- Invalid model output → repair retry, then grounded template option
- Feed rate-limited → per-source freshness dot, other feeds unaffected

Five induced-failure drills pass in CI.

WHO PAYS FOR THE DECISION LAYER

The buyers already own the dashboards.

GOVERNMENTS

National disaster agencies and emergency managers pre-positioning against a forecast, not a headline.

NGOS

Humanitarian logistics teams choosing where to stage supplies under uncertainty.

INSURERS

Cat-risk desks pricing exposure and modeling loss under intervention scenarios.

LOGISTICS

Supply chains re-routing around unfolding hazards with quantified lead time.

Situational-awareness spend is established and growing. CrisisCommand sells the layer above it: the simulation that turns awareness into a defensible decision.

BUILD STATUS

Shipped, end to end.

- **Monte Carlo engine** flood + earthquake kernels, batched tensors, p10–p90
- **Grounded LLM layer** 3-option generation, briefings, template fallback
- **Holographic globe UI** Modes A→B→C, escalation chart, zones on Earth
- **Live ingestion** real GDACS/USGS, on-GPU dedup, freshness
- **Hardening** quality ladder, 5 failure drills, honest fallbacks

NEXT

- **AMD GPU deployment** vLLM serving + evidence capture (runbook ready)
- **Full hazard ensemble** cyclone + wildfire kernels, dynamic weighting
- **ReliefWeb + tension feeds** richer signal beyond the two solid sources

153 backend tests · smoke test · failure drills, all green.

—— TEAM IMITASI

COMMERCIAL TOOLS SHOW YOU THE CRISIS.
THIS SIMULATES YOUR DECISION.

CrisisCommand: an AI war room, powered by AMD GPUs.

AMD GPU | ROCM | VLLM | FIREWORKS AI | PYTORCH | FASTAPI | REACT · THREE.JS